

Emphysema (panacinar)

- **α 1-antitrypsin**
 - Found in serum, tissue fluids, macrophages
 - Function: inhibits proteases (elastase) secreted by neutrophils during inflammation
- **Genetic deficiency**
 - Encoded on proteinase inhibitor (Pi) locus on chromosome 14
 - Polymorphic
 - Normal: M allele
 - Abnormal: Z allele
 - Homozygous ZZ (.012% of US population) variant has very low amounts of AAT → 80% develop panacinar emphysema, also associated with liver cirrhosis (worse if smoker)
- Any injury induced by smoking which increases activation and influx of neutrophils into lung leads to release of (local) proteases; in absence of alpha-1-antitrypsin  excessive digestion of elastic tissue

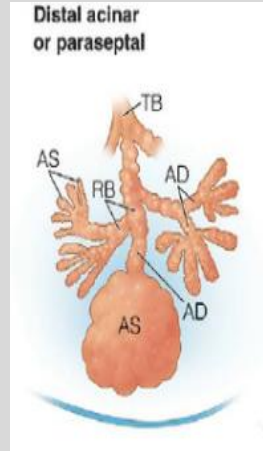
Emphysema

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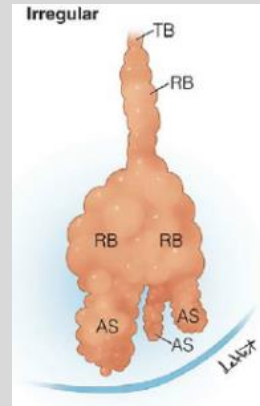
- **Paraseptal (distal acinar)**

- May be cause of spontaneous pneumothorax esp. in young males
- Proximal portion of acinus is normal, distal part predominantly involved
- Esp. adjacent to pleura (along lobular connective tissue septa and at margins of lobules)
- Occurs adjacent to areas of fibrosis, scarring, or atelectasis
- Usually more severe in upper half of the lungs
- Multiple enlarged airspaces, sometimes form cyst-like structures (bullae)



- **Irregular**

- Airspace enlargement with fibrosis
- Irregularly involved acinus
- Scarring usually present
- Usually clinically asymptomatic

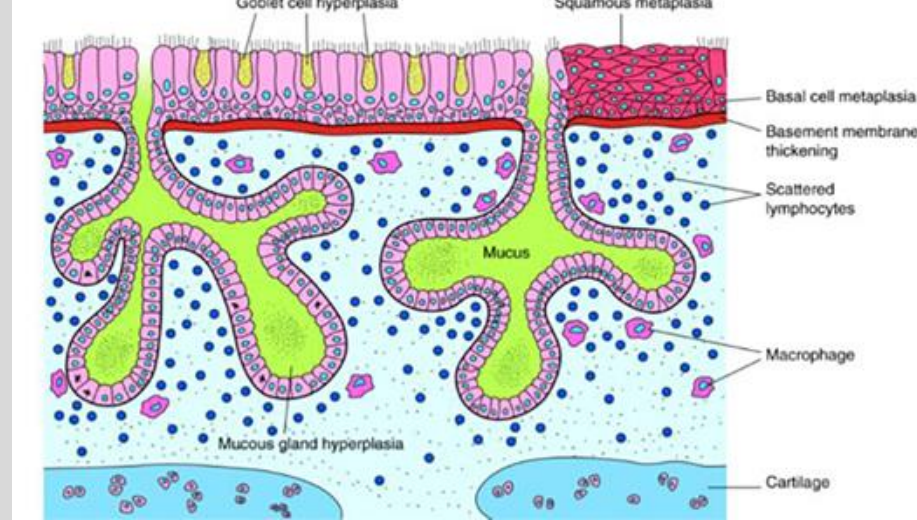


Emphysema: Pathogenesis

- **Toxic injury and inflammation:** cigarette smoke, etc. damages respiratory epithelium, causes inflammation, results in parenchymal destruction
 - Inflammatory mediators (e.g., leukotriene B₄, IL-8, TNF, etc.) are increased in affected parts of lung
 - Mediators released by resident epithelial cells and macrophages, attract inflammatory cells from circulation (chemotactic factors), amplify inflammation (proinflammatory cytokines), and cause structural changes (growth factors)
- **Protease-antiprotease imbalance:** Proteases released from inflammatory cells and epithelial cells break down connective tissue components
- **Oxidative stress:** Substances in tobacco smoke, alveolar damage, and inflammatory cells all oxidants, which may cause tissue damage, endothelial dysfunction, and inflammation
- **Infection:** Not initiating role but can exacerbate disease

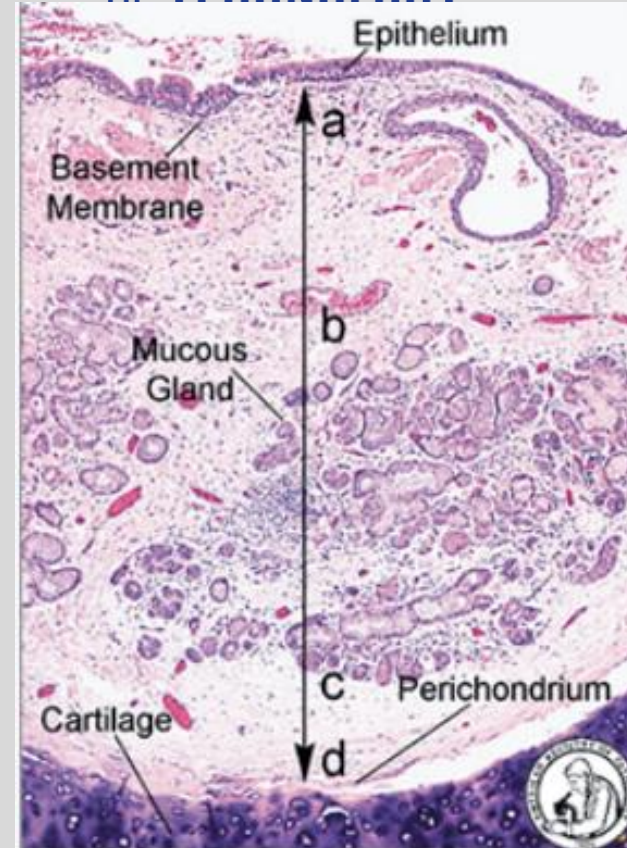
Chronic Bronchitis: Pathogenesis

- **Initiating factor:** exposure to noxious stimuli (tobacco smoke, dust from grain, cotton, silica)
- **Mucus hypersecretion**
 - **Hypertrophy of submucosal glands** – may involve inflammatory mediators (e.g., histamine, IL-13)
 - Large airways (trachea and bronchi)
 - Stimulated by proteases and metalloproteases
 - **Increased goblet cells**
 - Small airways (small bronchi and bronchioles)
 - Excessive mucus production → airway obstruction



Chronic Bronchitis: Morphology

- **Micro: Chronic inflammation** (lymphocytes, macrophages)
 - Thickening of bronchiolar wall secondary to smooth muscle hypertrophy, peribronchial fibrosis
 - **Increased goblet cells** (goblet cell hyperplasia)
 - **Enlargement (increase in size) of mucus glands of trachea and bronchi**
 - **Reid index – ratio of**
 - Thickness of mucous glands (b-c) over thickness of bronchial wall (a-d) between epithelium and cartilage
 - Normal 0.4; increased in chronic bronchitis
- **Narrowing of bronchioles**
 - **Mucus plugs, inflammation, fibrosis** – if severe: obliteration of lumen due to fibrosis (bronchiolitis obliterans)
- Bronchial epithelium may also show **squamous metaplasia** or dysplasia because of irritating and mutagenic effects of substances in tobacco smoke



Asthma

- **Chronic recurrent inflammatory disease of the airway**
 - Causes recurrent wheezing, breathlessness, chest tightness and cough
 - Bronchi & terminal bronchioles
 - Children > adults
 - Seasonal, exercise-induced, drug-induced, occupational, or in smokers
- **Increased airway responsiveness to many stimuli** leading to:
 - Episodic bronchoconstriction and airflow obstruction (at least **partly reversible**)
 - Inflammation of **bronchial walls**
 - Increased mucus secretion
- In all types: episodes of bronchospasm with diverse triggers (e.g., respiratory infections (especially viral infections), irritants (e.g., smoke, fumes), cold air, stress, and exercise

Asthma

- **Atopic (extrinsic)** – **allergen sensitization** and immune activation, often in patient with allergic rhinitis, or eczema and family history of asthma
 - Triggered by environmental allergens (dusts, pollens, cockroach or animal dander, and foods), mostly act in synergy with other proinflammatory environmental cofactors (especially respiratory viral infections)
 - Usually begins in childhood
 - **Th2 mediated IgE** response hypersensitivity (type 1)
 - Skin test results in wheal-and-flare reaction
 - May also be diagnosed based on **high total serum IgE levels** or evidence of allergen sensitization by serum radioallergosorbent tests (RASTs) - detect presence of IgE antibodies specific for individual allergens

Asthma

- **Non-atopic (intrinsic)** – no evidence of allergen sensitization, skin tests usually negative
 - Commonly triggered by viral respiratory tract infection (rhinovirus, parainfluenza virus, RSV)
 - Inhaled air pollutants (tobacco smoke, sulfur dioxide, ozone, nitrogen dioxide) may also contribute to chronic airway inflammation and hyperreactivity in some cases
 - Some attacks may be triggered by exposure to cold and exercise

Asthma

- **Drug-induced** – e.g., aspirin-sensitive asthma (uncommon) (also NSAIDS)
 - Asthmatic attacks and **urticaria**
 - Often occurs in patients with **recurrent rhinitis and nasal polyps**
 - Via inhibition of COX pathway: leads to decrease in PGE2, which inhibits enzymes that generate proinflammatory mediators (leukotrienes B4, C4, E4)
- **Occupational** – triggered by fumes, organic and chemical dusts, gas, or other **chemicals (e.g formaldehyde)**
 - Minute amounts induce attack, usually after repeated exposure
 - Various mechanisms including type I reactions, direct liberation of bronchoconstrictor substances, and hypersensitivity responses of unknown origin

Asthma: Pathogenesis

- Atopic asthma (most common form) caused by Th2-mediated IgE response to environmental allergens in genetically predisposed people
- **Airway inflammation** causes airway dysfunction by **release of potent inflammatory mediators** and **remodeling of airway wall**; with severe disease: **increased secretion of growth factors** (local) **induce mucous gland enlargement, smooth muscle proliferation, angiogenesis, and fibrosis.**
- **Fundamental abnormality: exaggerated Th2 response to normally harmless environmental antigens**
- Th2 cells secrete cytokines (e.g., IL-4) that promote inflammation and stimulate B cells to produce IgE and other antibodies
- Cytokines stimulate production of IgE; IL-5, (activates locally recruited eosinophils); and IL-13 (stimulates mucus secretion from bronchial submucosal glands, promotes IgE production by B cells)
- T cells and epithelial cells secrete chemokines that recruit more T cells and eosinophils, causing exacerbation
- **IgE binds to Fc receptors on submucosal mast cells**; repeat exposure to allergen triggers mast cells to release granule contents and produce cytokines and other mediators, together inducing the early-phase (immediate hypersensitivity) and late-phase reactions

Asthma: Pathogenesis (continued)

- **Early-phase reaction:**
bronchoconstriction, increased mucus production, variable degrees of vasodilation, and increased vascular permeability
- Bronchoconstriction triggered by direct stimulation of subepithelial vagal (parasympathetic) receptors by both central and local reflexes (triggered by mediators produced by mast cells and other cells)
- **Late-phase reaction: recruitment of leukocytes, especially eosinophils, neutrophils, and more T cells (Th2 cells dominant T-cell type; Th17 (IL-17 producing) cells recruit neutrophils**
- **Mediators in bronchospasm** (supported by drug therapy success) include
 - (1) **leukotrienes C₄, D₄, and E₄: cause prolonged bronchoconstriction, increased vascular permeability, increased mucus secretion**
 - (2) **acetylcholine**, released from intrapulmonary parasympathetic nerves: **causes airway smooth muscle constriction by directly stimulating muscarinic receptors**
 - (3) **IL-5**, associated with **peripheral blood eosinophilia**
 - (4) **galectin-10 (GAL10)**, released from eosinophils and forms **Charcot-Leyden crystals, strong inducers of inflammation and mucus production**

Asthma – Environmental Factors

- Mostly affects urban populations
- Cities have more airborne pollutants which may be allergens initiating Th2 response
- Limited exposure of very young children to certain antigens (**hygiene hypothesis** – microbial exposure during early life may reduce later incidence of allergic diseases)
- Infections can cause acute exacerbations
- Young children who develop viral respiratory tract infections (e.g., rhinovirus type C, RSV) have 10 to 30-fold increase of developing persistent and/or severe asthma

Asthma - Clinical

- Classic acute asthmatic attack lasts up to several hours
- **Chest tightness, dyspnea, expiratory wheezing, and coughing (with or without sputum production)** – for some patients constantly present at low level
- Diagnosis based on demonstration of **increase in airflow obstruction (from baseline levels); difficulty with exhalation (prolonged expiration, wheeze)**
- **In atopic asthma: also eosinophilia in peripheral blood and eosinophils, Curschmann spirals, and Charcot-Leyden crystals in sputum**
- **Pulsus paradoxus (systolic bp drop greater than 10mmHg during inspiration)**
 - In severe asthma attack, parts of airways tighten and swell, lungs begin to overinflate in response
 - Puts pressure on veins carrying unoxygenated blood from heart to lungs; leads to back up of blood in right ventricle causing extra pressure on right side which presses against left side of heart
 - Increased negative pressure in lungs puts extra pressure on left ventricle
- Increased AP diameter during attacks
- **Status asthmaticus** – severe bronchoconstriction that does not respond to drugs that usually abort acute attack
 - Most severe form of asthma, lasts days to weeks, can be fatal

Asthma - Clinical

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- **Labs**
 - Initial: **respiratory alkalosis**
 - Can progress to **respiratory acidosis** if bronchospasm not relieved
 - Intubation and mechanical ventilation indicated if patient has a normal pH or respiratory acidosis
 - Eosinophilia
 - Positive skin test for allergens
- **Treatment** based on severity
 - Include bronchodilators, glucocorticoids, and leukotriene antagonists
 - Severe asthma: new biologic therapies targeting inflammatory mediators such as IL-5 blocking antibodies

Bronchiectasis

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- Destruction of smooth muscle and elastic tissue by inflammation from persistent or severe infections
- Leads to **permanent dilation** of bronchi and bronchioles.
- Associated with **chronic necrotizing infections**
 - Obstruction or impaired clearing → pooling of secretions distal to the obstruction and inflammation → necrosis, fibrosis, dilation



Bronchiectasis

- Now uncommon, but may develop in association with:
 - Congenital or hereditary conditions that predispose to chronic infections (**cystic fibrosis**, intralobar sequestration of the lung, immunodeficiency states, **primary ciliary dyskinesia**, and **Kartagener syndrome**)
 - **Severe necrotizing pneumonia** caused by bacteria, viruses, or fungi
 - **Bronchial obstruction**, due to tumor, foreign body, or mucus impaction
 - **Immune disorders** (rheumatoid arthritis, systemic lupus erythematosus, inflammatory bowel disease, and posttransplant (chronic rejection after lung transplant and chronic graft-versus-host disease after hematopoietic stem cell transplantation)
- Up to 50% of cases idiopathic; appear to be dysfunctional host immunity to infectious agents leading to chronic inflammation

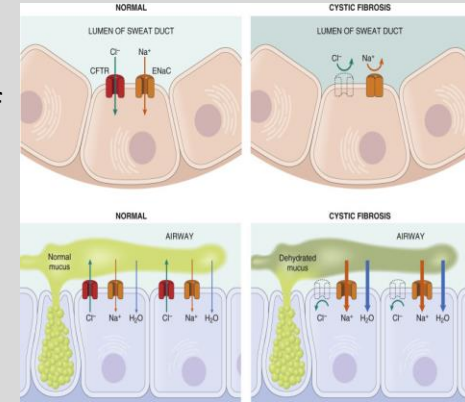
Bronchiectasis: Pathogenesis: Obstruction and Infection

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- **Congenital or hereditary conditions**

- **Cystic Fibrosis** (most common cause in US) (AR)
 - Defect in CFTR for chloride ions (defect in ion transport) → dehydrated thick secretions: affect mucociliary clearance and lead to airway destruction
 - Chronic bacterial infections: cause widespread damage to airway walls
 - Bronchi markedly dilated, smaller bronchioles progressively obliterated because of fibrosis (bronchiolitis obliterans)
 - Associated with salty-tasting skin and meconium ileus
- **Primary ciliary dyskinesia** (AR)
 - Mutations causing ciliary dysfunction due to ciliary motor proteins' defects (mutations in dynein arm in cilia)
 - Poorly functioning or absent dynein arm in cilia prevents mucociliary clearance → retention of secretions & recurrent infection
- **Kartagener** (AR) – present in about half of patients with primary ciliary dyskinesia
 - Ciliary disorder (ciliary function needed during embryogenesis for proper rotation of organs in chest and abdomen)
 - Associated with **situs inversus and male infertility** (result of sperm dysmotility)



Bronchiectasis

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- **Infections**
 - **TB** (most common cause worldwide)
 - **Allergic bronchopulmonary aspergillosis** (may occur in asthma or cystic fibrosis)
 - Hyperimmune reaction to *Aspergillus fumigatus* – sensitization leads to activation of Th2 helper T cells that release cytokines which recruit eosinophils and other wbc's
 - High serum IgE levels, serum antibodies to *Aspergillus*, intense airway inflammation with eosinophils, and formation of mucus plugs
 - Fungal hyphae may be seen on special stains within mucoinflammatory contents of dilated segmental bronchi; later, may infiltrate bronchial wall
- **Bronchial obstruction**
 - Tumor
 - Foreign bodies
 - Mucus impaction
- **Immune disorders**
 - RA, SLE, IBD, post-transplant

Bronchiectasis: Morphology

- **Gross:**
 - Usually affects **lower lobes bilaterally**, most severe in **more distal bronchi and bronchioles**
 - **Airways dilated**, sometimes up to four times normal size; bronchi and bronchioles can be followed almost to pleural
 - Cut surface of lung: dilated bronchi appear cystic, filled with mucopurulent secretions
- **Microscopic**
 - **Acute and chronic inflammatory exudate** in walls of bronchi/bronchioles
 - Desquamation of epithelial lining; squamous metaplasia of remaining epithelium because of inflammation, exacerbating mucociliary clearance
 - Areas of **necrotizing ulceration**
 - Bronchial wall destruction may lead to abscess formation
 - **Chronic cases may show fibrosis of bronchial and bronchiolar walls which leads to partial or total obliteration of lumens**
 - **Mixed flora** of bacteria usually cultured from bronchi
 - *H. influenza* found in half and *P. aeruginosa* in 12-30% of sputum cultures

Bronchiectasis: Clinical

- **Severe, persistent cough with abundant, foul smelling**, sometimes bloody, **sputum**
 - **Paroxysms of cough common in morning** due to drainage of collected pools of pus/secretions as patient changes positions
- **Dyspnea, orthopnea**; if severe – hemoptysis
- Obstructive respiratory insufficiency may lead to marked dyspnea, cyanosis
- Treatment with better antibiotics and physical therapy improve prognosis; historically cor pulmonale, brain abscesses, etc. no longer as frequent

Clinical Term	Anatomic Site	Major Pathologic Changes	Etiology	Signs/Symptoms
Chronic bronchitis	Bronchus	Mucous gland hyperplasia, hypersecretion	Tobacco smoke, air pollutants	Cough, sputum production
Bronchiectasis	Bronchus	Airway dilation and scarring	Persistent or severe infections	Cough, purulent sputum, fever
Asthma	Bronchus	Smooth muscle hyperplasia, excess mucus, inflammation	Immunologic or undefined causes	Episodic wheezing, cough, dyspnea
Emphysema	Acinus	Airspace enlargement; wall destruction	Tobacco smoke	Dyspnea

Question

A 12-year-old girl has a **history of coughing and wheezing and repeated attacks of difficulty breathing, especially in the spring**. During these episodes she is afebrile. PE reveals **hyperresonant lungs, and CXR shows increased lucency of all lung fields. Laboratory tests show an elevated serum IgE level and peripheral blood eosinophilia**. A sputum sample examined microscopically also has **increased numbers of eosinophils**. Which of the following histologic features is most likely to characterize the lung in this patient's acute condition?

- (A) Dilation of respiratory bronchiole and distention of alveoli
- (B) Dilation of bronchi with inflammatory destruction of walls
- (C) Interstitial and alveolar edema with presence of hyaline membranes that line alveoli
- (D) Thickening of bronchial epithelial basement membrane and hypertrophy of bronchial smooth muscle
- (E) Patchy areas of consolidation surrounding bronchioles and neutrophilic exudate in affected alveoli

Question

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A 35-year-old woman has had **multiple bouts of severe necrotizing pneumonia with Haemophilus influenzae, Staphylococcus aureus, Pseudomonas aeruginosa, and Serratia marcescens cultured from her sputum since childhood.** She now has for weeks at a time a **cough productive of large amounts of purulent sputum.** PE reveals dullness to percussion with decreased breath sounds over the right mid to lower lung fields. A **bronchogram shows marked dilation of right lower lobe bronchi.** Which of the following mechanisms is the most likely cause of airspace dilation in this patient?

- (A) Unopposed action of neutrophil-derived elastase
- (B) Congenital weakness of supporting structures of the bronchial wall
- (C) Diffuse alveolar damage
- (D) Destruction of bronchial walls by recurrent inflammation
- (E) Damage to bronchial mucosa by major basic protein of eosinophils

Question

A 20-year-old, previously healthy man is jogging one morning when he trips and falls to the ground. He suddenly becomes markedly short of breath. His jogging partner brings him to the ED where auscultation of the lungs reveals **no breath sounds audible over the right chest. CXR shows shift of the mediastinum from right to left. A right chest tube is inserted, and air rushes out.** Which of the following types of obstructive lung disease is the most likely diagnosis?

- (A) Asthma
- (B) Bronchiectasis
- (C) Centriacinar emphysema
- (D) Chronic bronchitis
- (E) Distal acinar emphysema

Question

A 50-year-old man comes to the physician with **gradually increasing dyspnea and a 4-kg weight loss over the past 2 years**. He admits to **smoking two packs of cigarettes** per day for 20 years, but he has not smoked for the past year. PE shows an increase in the anteroposterior diameter of the chest (**“barrel chest”**). Auscultation of the chest shows decreased lung sounds. A chest radiograph shows bilateral hyperlucent lungs, especially marked in the upper lobes. Pulmonary function tests show that the FEV1 is markedly decreased, but the FVC is normal, and **FEV1/FVC ratio is decreased**. Which of the following is most likely to contribute to the pathogenesis of his disease?

- (A) Impaired hepatic release of α 1-antitrypsin
- (B) Release of elastase from neutrophils
- (C) Abnormal epithelial cell chloride ion transport
- (D) Decreased ciliary motility with irregular dynein arms
- (E) Macrophage recruitment and release of interferon- γ